3.9.3 Proposed mitigation measures

The mitigation measures proposed for the four study areas are informed by the Catchment Restoration Opportunity Mapping Support System (CROM DSS). The emphasis is here again placed on the assumption that, should landslides occur, these would be shallow landslides leading mostly to soil erosion. This assumption provides the rationale for the application of CROM DSS measures. The Catchment Restoration Opportunities Matrix presented in the Table 17 below summarizes the different erosion control measures according to the slopes, soil depth categories and geology.

Table 17: **CROM matrix of slope / soil depth classes and alternative land use options**

Soil depth / Land slope	> 0.5 m	< 0.5 m
1: (0-6%)	**Class I** ▪ Agroforestry + contour ploughing + alley cropping with grass strips	**Class VI** ▪ Agroforestry + contour ploughing + alley cropping with grass strips ▪ Forestation where soil depth is too limited and unsuitable for crops. ▪ Perennial crops, coffee, tea, banana, fruit trees
2: (6 - 16%)	**Class II** ▪ Progressive terraces (reinforced by agroforestry hedges and grass strips) ▪ Perennial crops, coffee, tea, banana, fruit trees	**Class VII-a** ▪ Progressive terraces (reinforced by agroforestry hedges and grass strips) ▪ Perennial crops, coffee, tea, banana, fruit trees ▪ Forestation where soil depth is too limited and unsuitable for crops
3: (16 - 40%)	**Class III** ▪ Bench terraces (option only in case of suitable, stable parent material / geology; avoid introducing landslide risks) ▪ Progressive terraces (reinforced by agroforestry hedges and grass strips) ▪ Perennial crops, coffee, tea, banana, fruit trees	**Class VII-b** ▪ Progressive terraces (reinforced by agroforestry hedges and grass strips) ▪ Forestation where soil depth is too limited and unsuitable for crops ▪ Perennial crops, coffee, tea, banana, fruit trees
4: (40- 60%)	**Class IV** ▪ Narrow cut terraces (option only in case of suitable, stable parent material / geology; avoid introducing landslide risks) ▪ Progressive terraces (reinforced by agroforestry hedges and grass strips) ▪ Forestation (Biological measures) ▪ Perennial crops, coffee, tea, banana, fruit trees	**Class VIII-a** ▪ Forestation (Biological measures) + trenches / ditches
5: (> 60)	**Class V** ▪ Forestation (Biological measures) + trenches / ditches ▪ Perennial crops, coffee, tea, banana, fruit trees	**Class VIII-b** ▪ Natural vegetation

In line with the existing land use in each of the sub-catchments, the following mitigation measures are proposed: